

Information Technology, Information Security, Social Media and Surveillance

ENGR49580D Technology, Society and Sustainability

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- Network topologies
 - Circuit Switched
 - Packet Routed
- Network Surveillance and Encryption
- Social Media
 - Who pays for ‘the internet’?
- Net Neutrality
- Surveillance

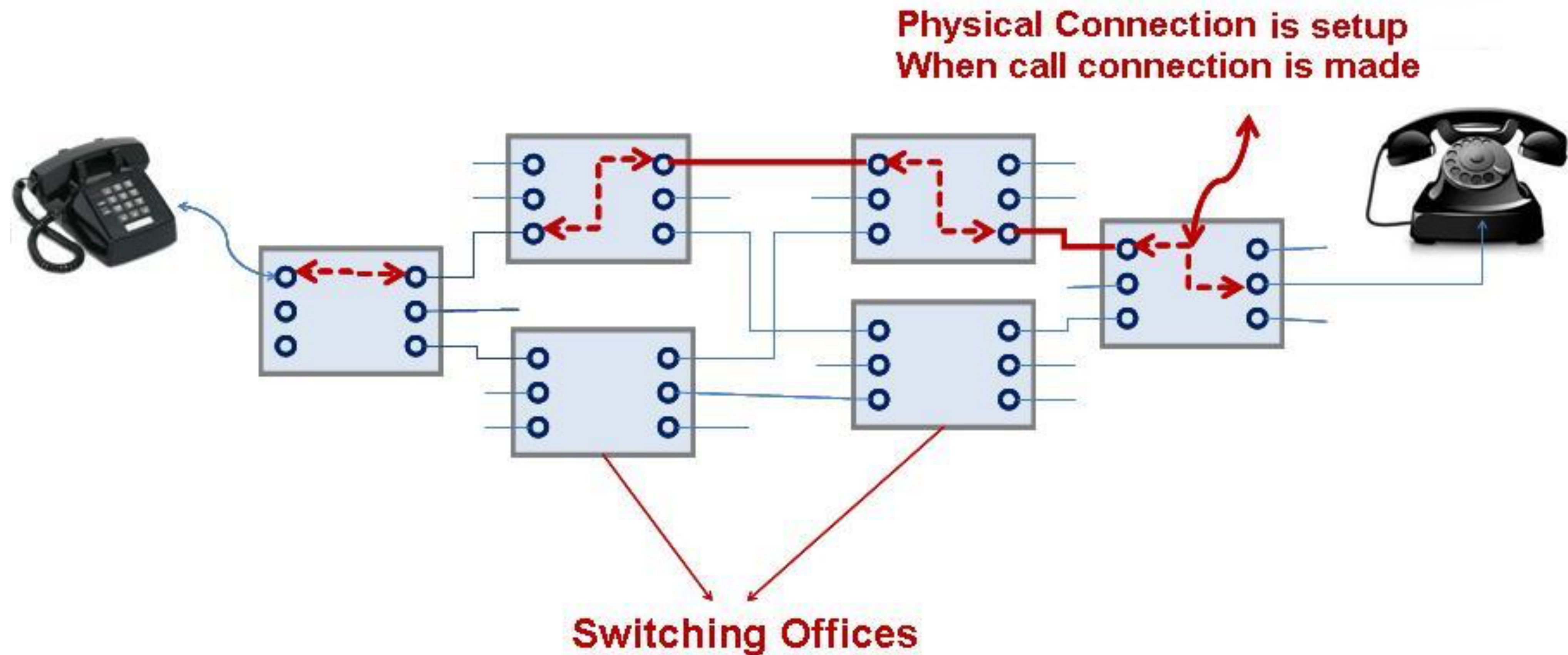
Network Topologies

The importance of network design

- When information is transmitted from one location to another, the route that it takes makes it vulnerable
- Information may be:
 - Monitored
 - Altered
 - Delayed
 - Stopped

Network Topologies

Circuit Switched Networks



Network Topologies

Circuit Switched Networks

- A connection is established between nodes by switching interconnecting lines to establish a link
- Multiplexing (wire sharing) may be used to increase the usable capacity of resources
- Bandwidth is dedicated to the connection
- Since the connection is fixed, it is possible to assess the security of the connection
- Typical of monopoly telephone systems

Network Topologies

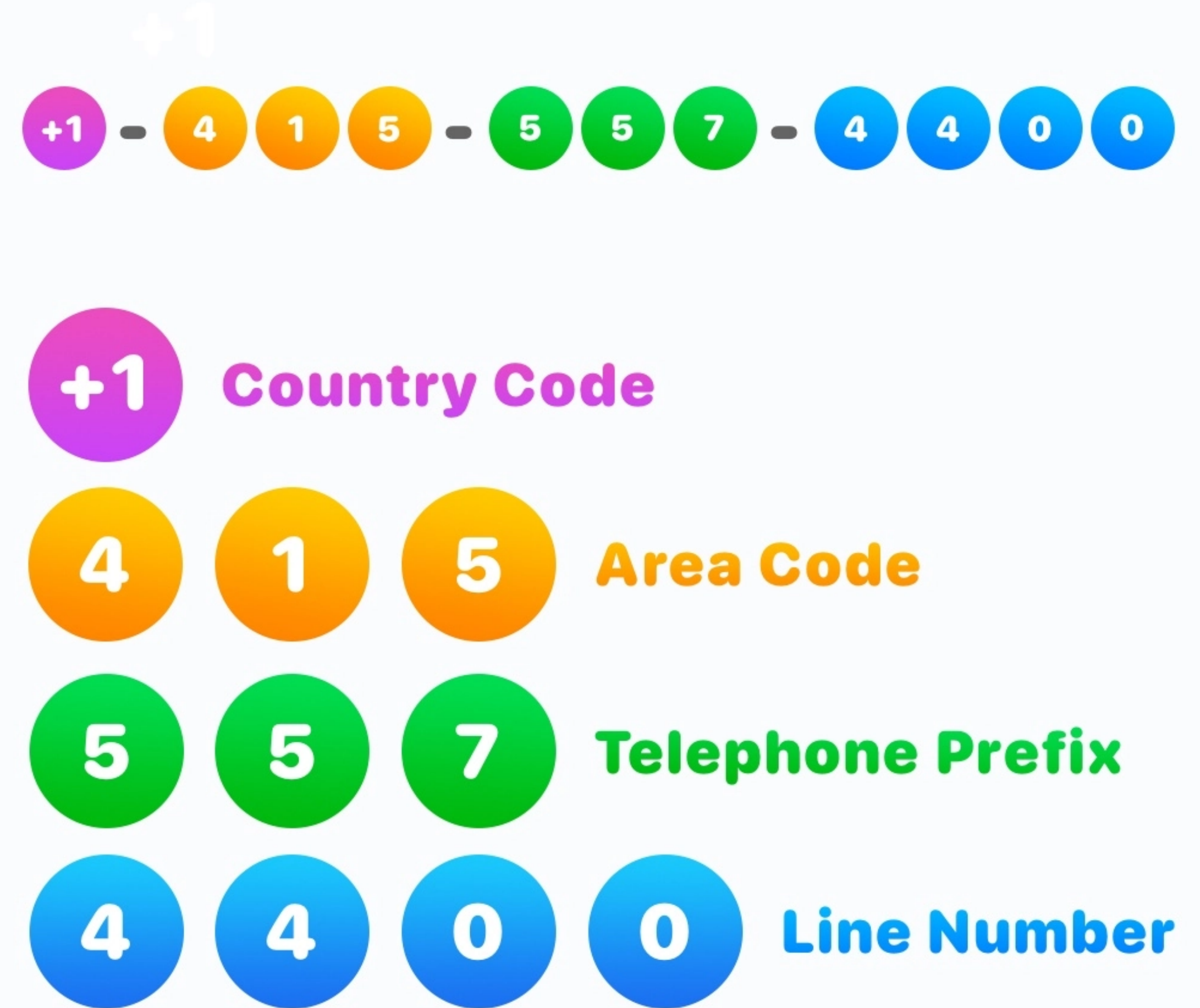
Circuit Switched Networks - Advantages

- Dedicated bandwidth assures consistent connection quality
- Since the connection route is known, it is possible to establish trust
 - Under the monopoly telephone systems only a few network operators (often only 1) were involved in carrying a connection

Network Topologies

Circuit Switched Networks - The Monopoly Telephone System

- Telephone numbers were the encoded routing information for the point to point connection
- The area code represented a large geographic region
- The prefix represented the local exchange where the line was terminated
 - Each exchange served 1 or more prefixes



Network Topologies

Circuit Switched Networks - Surveillance Techniques

- Connection / Call records
 - These would be available only from from the network operator (if logged)
 - Pay-per-use connections (such as long distance calls) are typically logged for billing purposes
 - Records of non-pay-per-use connections may be available
 - Called Local Usage Details (LUDs) for telephone systems
- Network operators will typically only release this information to law enforcement

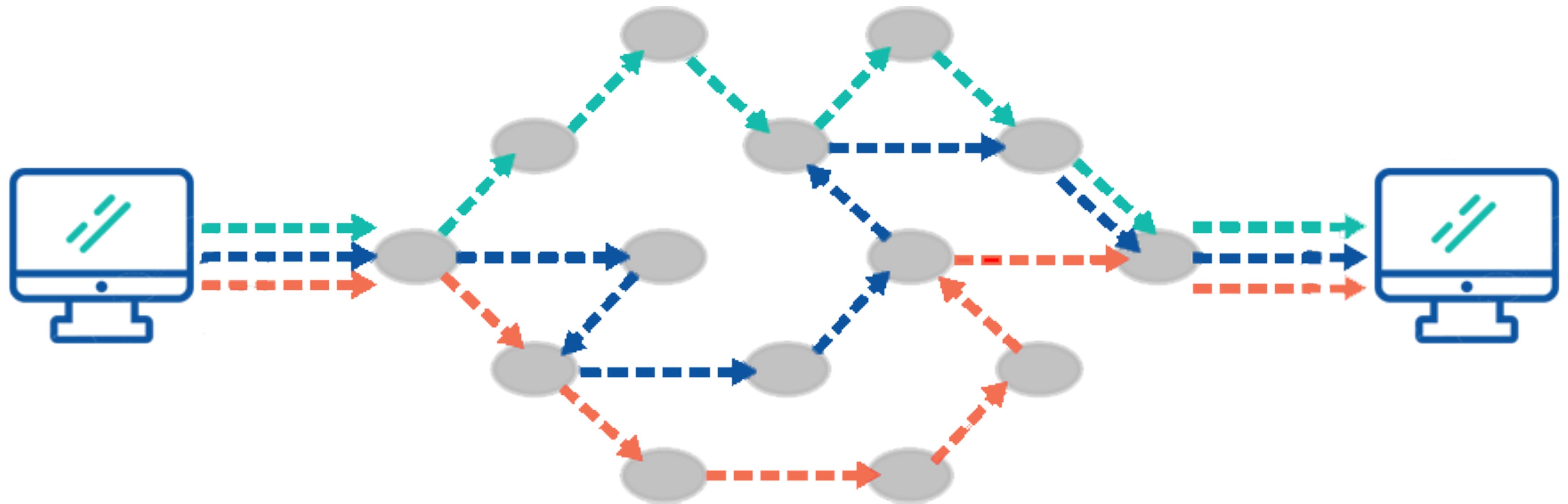
Network Topologies

Circuit Switched Networks - Surveillance Techniques

- Wire Tap
 - A physical device is installed on the network circuit to record or retransmit data
 - May be located anywhere on the circuit
 - Legal (court ordered) wire taps are typically installed by the network operator at the exchange
 - Unauthorized wire taps may be located anywhere there is physical access to the circuit such as a junction box or exposed cable

Network Topologies

Packet Routed Networks



Network Topologies

Packet Routed Networks

- No direct connection is established between communicating nodes
- Information is split into self contained packages (packets) that also contain sender and receiver information
- Routers forward packets along routes to their destination
- Resources are not tied up with single connections
- Bandwidth is shared between all network users
- It is not always possible to determine packet route before transmission

Network Topologies

Packet Routed Networks - Routing Strategies

- Static Routes - between known points on a stable network
- Dynamic Routing - the router will use an algorithm to determine the best route for each package through the network
 - Requires the router to have a complete and accurate map of the network (routing table)
- Next Hop Routing - the router sends the packet to another router on a path towards the destination
 - Routing Tables only need to contain information about directly connected networks
 - The packet path is not knowable at the time of transmission

Network Topologies

Packet Routed Networks - Advantages

- Bandwidth is shared between all network users
 - Less idle bandwidth
 - Lower infrastructure cost

Network Topologies

Packet Routed Networks - Disadvantages

- Complex, dynamic networks (such as the public internet) are typically Next Hop Routed
- Since the packet route may not be known at the time of transmission, trust is difficult to establish

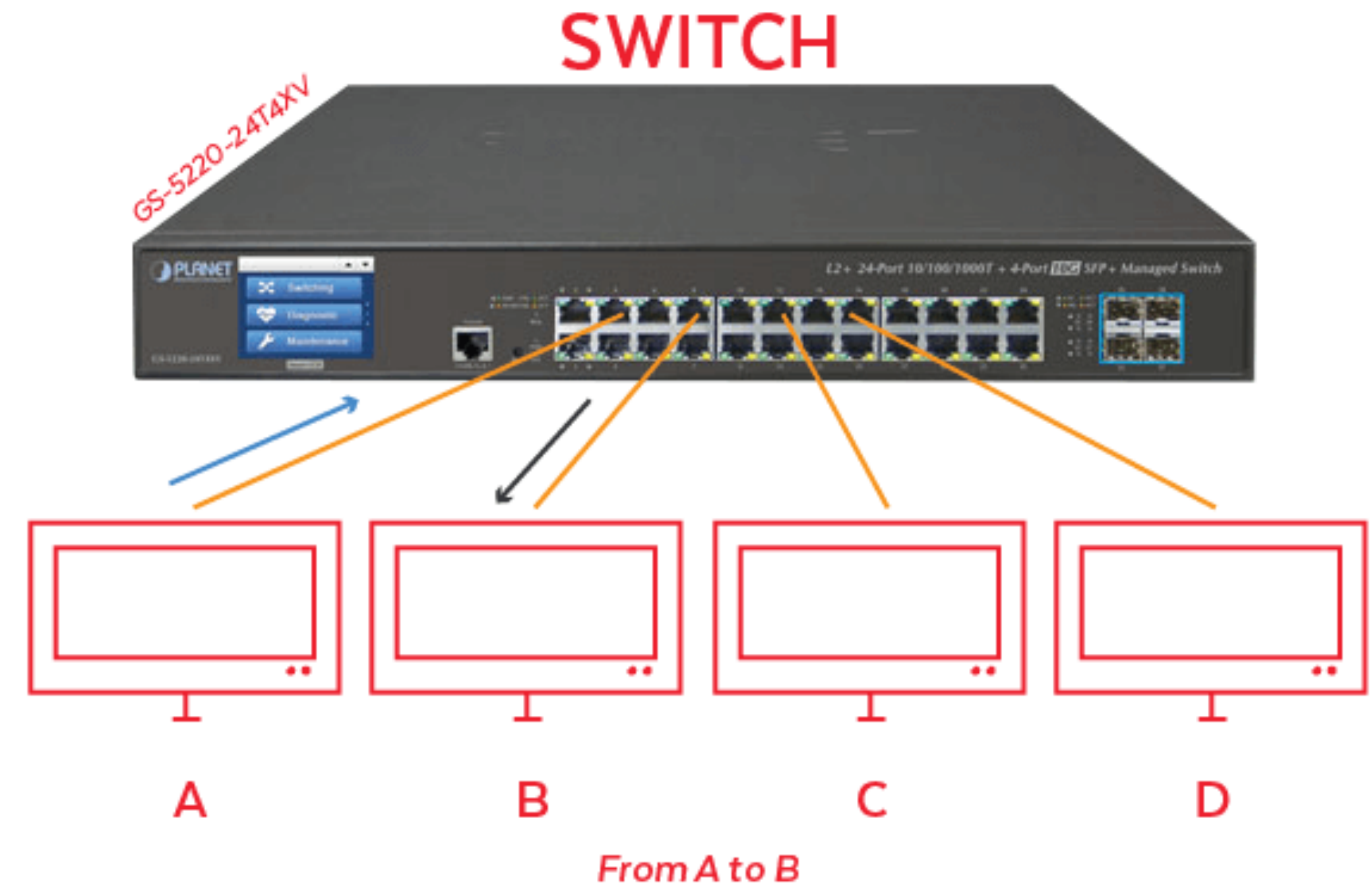
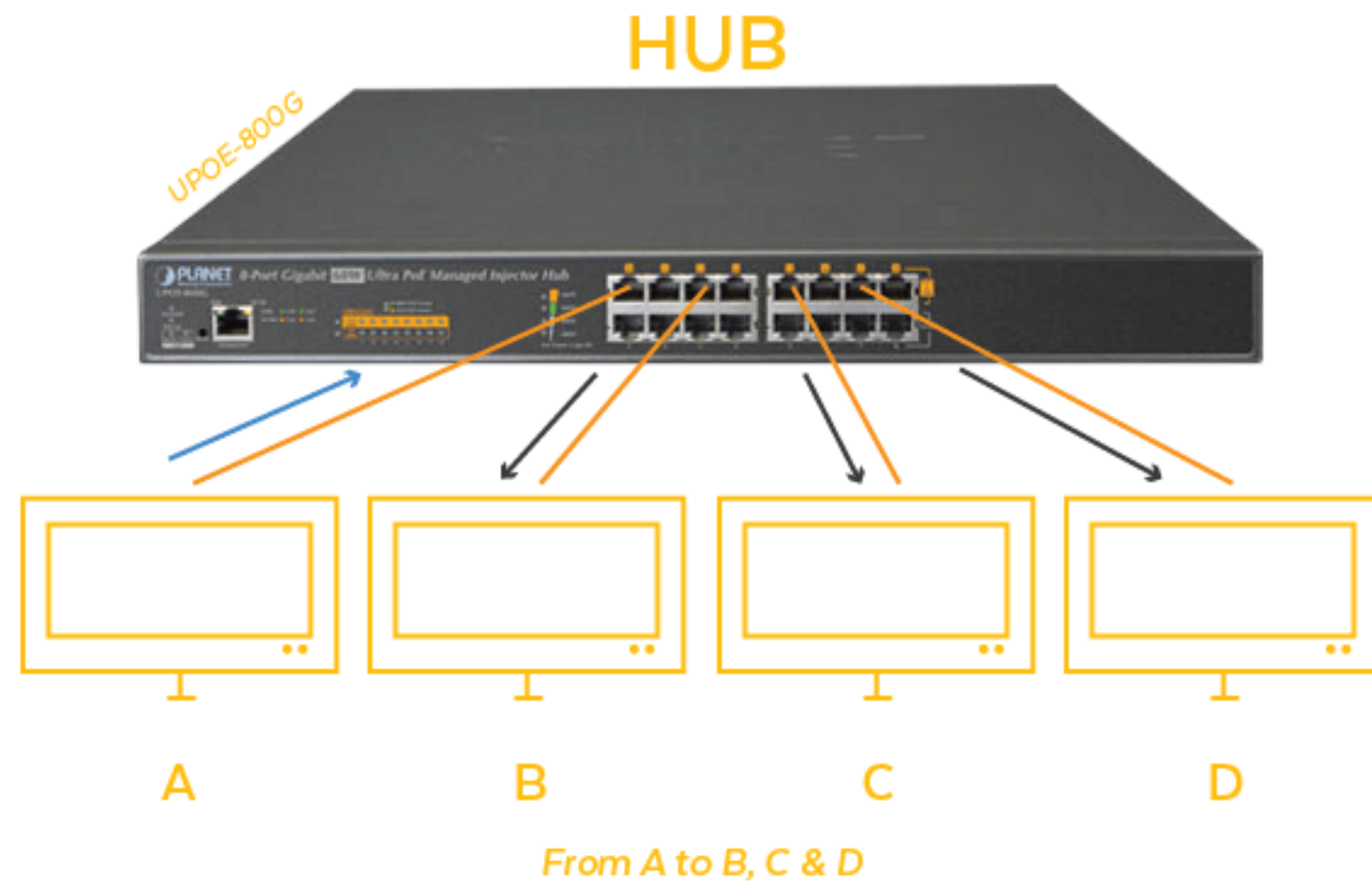
Network Topologies

Packet Routed Networks - Surveillance Techniques

- Packet sniffing
 - Specialized software reads content of packets as they are passed along the network
 - The packet must be transmitted to (through) the system running the packet sniffer
 - Networks that use hubs instead of switches are more vulnerable
 - Encrypted packets still contain unencrypted routing information (where the packet is coming from / going to)

Network Topologies

Hubs vs Switches



Network Topologies

Packet Routed Networks - Surveillance Techniques

- Spoofing
 - A device impersonates a trusted network device with the intention of having information routed to it
 - This may be done for:
 - Packet sniffing
 - User Fraud (phishing)
 - Information modification / deletion / delay

Encryption

- On public networks, the preferred approach to security is encryption
- Encryption essentially rearranges data, making it appear random, and requiring decryption to make it readable again
- Two approaches to encryption
 - Symmetrical: the same key is used to encrypt and decrypt data
 - This key must be kept secure
 - Asymmetrical: separate keys are used to encrypt and decrypt data
 - Keys may be shared with less risk to data security

Encryption Strategies

- Server - Client
 - Packets are encrypted by both the client and server
 - Use case is secure transactions such as banking
- End-to-End
 - Packets are encrypted by one user and decrypted by another without the server having the ability to decrypt them
 - Use case is messaging (eg WhatsApp)

“If you’re not paying for the product, you are the product.”

Attribution Unknown

Social Media

- Social Media and Search companies have become a target for their extensive use of advertising sales as a revenue stream
- Technology has allowed these companies to build individual consumer profiles to very effectively target ads

Social Media

Ad Supported Services

- Advertising supported services are not new and have used consumer profiles to target ads for a long time:
 - Newspapers
 - Magazines
 - Television
 - Transit Systems
- Unlike many free to use social media / search platforms, these traditional media are often paid for twice

Social Media

Privacy Concerns

- Cross platform tracking
 - Use of cookies, tracking pixels, or other technology to track users across multiple sites / apps
- Meaningful consent for tracking / cookies etc.
- Sale / release of user data to 3rd parties (including NSA)

Net Neutrality

Principle

- Internet Service Providers (ISPs) should treat all internet traffic equally
- Provide access to all sites, content and applications at the same speed

Net Neutrality

Actions against net neutrality

- Traffic Shaping / QoS
 - Prioritizing certain applications or throttling others
 - Typically done under the guise of improving user experience
- Censorship
 - Blocking certain content / domains
 - Typically done due to government mandate

Net Neutrality

Actions against net neutrality

- Zero Rating
 - Not charging users on pay-per-use accounts for certain types of traffic
- May be part of a fee structure such as a 'flat rate streaming' plan
- May favour services owned by or partnered with the ISP

+ Smart Net
Oferta da 1ª mensalidade de uma Smart Net com 10GB/mês adicionais ⁽¹⁾

MESSAGING	SOCIAL	VIDEO
		
€4,99/mês €6,99/mês 1 mês grátis Aderir	€4,99/mês €6,99/mês 1 mês grátis Aderir	€4,99/mês €6,99/mês 1 mês grátis Aderir
MUSIC	EMAIL&CLOUD	MEO
		
€4,99/mês €6,99/mês 1 mês grátis Aderir	€4,99/mês €6,99/mês 1 mês grátis Aderir	Tráfego grátis para apps MEO já incluído no seu tarifário

Net Neutrality

Actions against net neutrality

- Server-Pay deals
 - Content providers (websites, social media, streaming services) pay ISPs for preferred access to their network
 - Preferred access means faster connections to services that are able to pay

Surveillance

- Many governments acknowledge monitoring social medial activity
 - Statutory limits to this surveillance are in place in many jurisdictions
 - Limits vary significantly between jurisdictions

Surveillance

Backdoors

- A backdoor is a feature built in to a piece of software that allows the author or a third party to access the software on a user's device
- Skype (an end-to-end encrypted platform) had a back door allowing communications to be monitored at the end user's pc (by the NSA according to the Snowden Leaks)
- In 2016 Apple was ordered to create a version of IOS with a back door to allow the FBI to access the phone of a mass shooting suspect
 - Apple fought the order and the FBI withdrew the request